Excess Returns of the US Manufacturing Index: A Fama-French Five-Factor Model Analysis

Technical Appendix

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A.1 Data and Variables

Monthly data covering the period from July 1964 to August 2022 were used in the analysis.

Sources

- Kenneth French Data Library
- Fama-French Five-Factor dataset
- US Manufacturing Index returns

Dependent Variable

- **Manuf** – excess returns of the US Manufacturing Index

Independent Variables

- **Mkt-RF** – market excess return
- **SMB** – Small Minus Big (size factor)
- **HML** – High Minus Low (value factor)
- **RMW** – Robust Minus Weak (profitability factor)
- **CMA** – Conservative Minus Aggressive (investment factor)
- **RF** – risk-free rate

A.2 Econometric Methodology

The analysis was conducted using Ordinary Least Squares (OLS) regression models in R.

Due to violations of classical assumptions in the full multivariate specification, the study additionally estimated separate univariate regressions for each explanatory factor.

Software and Packages

The analysis was conducted in R using packages including:

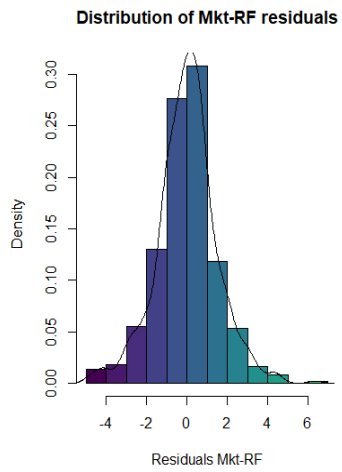
- readxl
- lmtest
- tseries
- car
- orcutt
- viridis

Diagnostic Tests Performed	Purpose	Result
F-test	Overall model significance	Most variables significant
Jarque-Bera test	Normality of residuals	Minor violations for Mkt-RF
Breusch-Pagan test	Heteroscedasticity	RF showed heteroscedasticity
Durbin-Watson test	Autocorrelation	No significant autocorrelation
VIF statistics	Multicollinearity	No serious multicollinearity detected

A.3 Key Figures from R Analysis

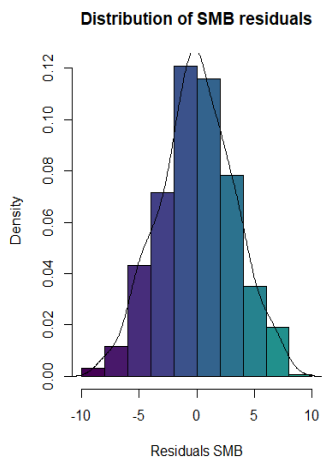
Below are selected outputs generated in R.

Figure 1. Distribution of Mkt-RF Residuals



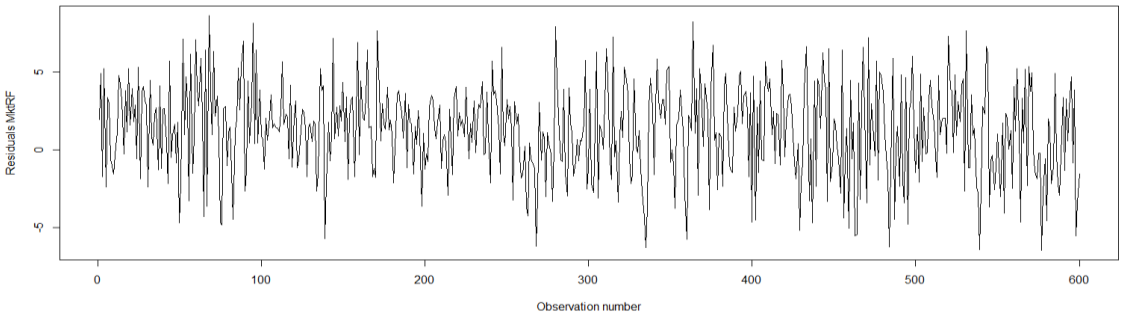
The histogram indicates an approximately bell-shaped distribution of residuals, although minor deviations from normality are visible due to the volatility of market returns.

Figure 2. Distribution of SMB Residuals



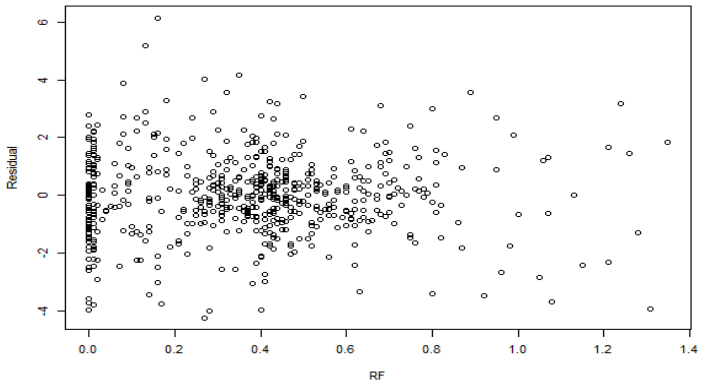
Residuals for the SMB model display a relatively symmetric distribution and visually satisfy the normality assumption.

Figure 3. Residuals of the Mkt-RF Model Over Time



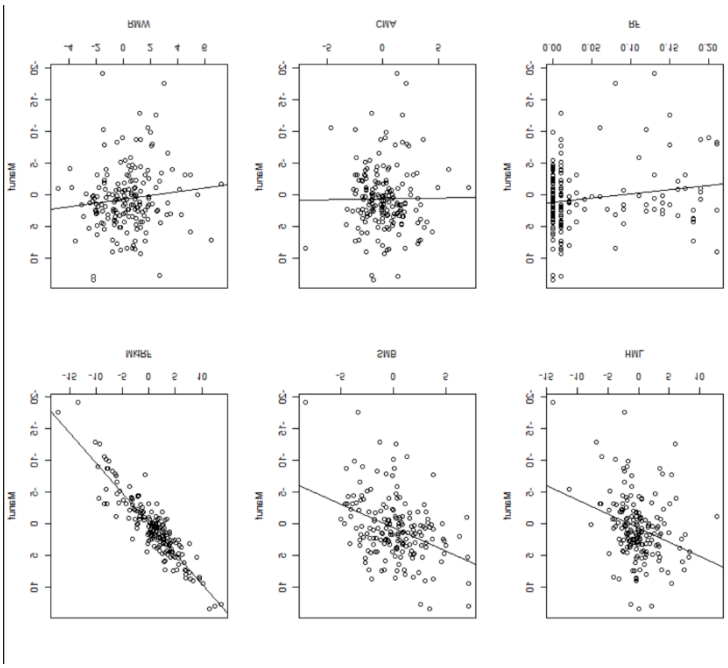
The residual series fluctuates around zero without a clear systematic trend, suggesting no major autocorrelation problems. This interpretation is consistent with the Durbin-Watson statistics reported in the analysis.

Figure 4. Residuals Against RF Values



The residual plot against RF values illustrates changing variance across observations, supporting the Breusch-Pagan test results indicating heteroscedasticity in the RF specification.

Figure 5. Scatterplots of Manufacturing Excess Returns Against Fama-French Factors



The figure illustrates the relationship between manufacturing excess returns and each explanatory variable included in the Fama-French framework. The strongest positive linear relationship is observed for Mkt-RF, while the remaining factors display substantially weaker explanatory power.

A.4 Selected Regression Results

Variable	Coefficient	Interpretation
Mkt-RF	Positive, strong	Main driver of manufacturing returns
SMB	Weak positive	Limited explanatory power
HML	Weak negative	Small value effect
RMW	Weak negative	Limited profitability effect
CMA	Weak negative	Small investment effect
RF	Very weak	Excluded due to heteroscedasticity

The strongest explanatory variable was Mkt-RF, accounting for approximately 79% of variation in manufacturing index excess returns. Other Fama-French factors showed substantially weaker explanatory power.

A.5 Limitations

Several limitations should be acknowledged:

- use of separate univariate regressions instead of a fully corrected multivariate specification,
- potential omitted variable bias,
- sensitivity of normality tests in large datasets,
- structural breaks across different economic periods between 1964 and 2022.